LEDGER MVP

GitHub Issue List & Sprint Plan

4 sprints × 2 weeks each = 8-week MVP build

Total: 26 issues organized by dependency and priority

Sprint	Duration	Theme	Issues
Sprint 1	Weeks 1–2	Data Foundation & Ingestion	7 issues
Sprint 2	Weeks 3–4	Feature Engineering & Credit Policy	7 issues
Sprint 3	Weeks 5–6	Shadow ML & Decisioning Engine	6 issues
Sprint 4	Weeks 7–8	Monitoring, Demo UI & Documentation	6 issues

Sprint 1: Data Foundation & Ingestion

Goal: Synthetic data generated, ingestion pipeline working, reconciliation tested.

#	Title	Labels	Pri	Size	Description & Acceptance Criteria
1	Project scaffolding & config	infra	P0	S	Create repo structure (10 dirs), config.py with all thresholds, requirements.txt, README. AC: • pip install works • config.py importable • All dirs exist with __init__.py
2	Synthetic merchant generator	data	P0	M	Implement generate_merchants() — 200 merchants, EUR 300k–5M GMV, 5% bad, KvK da AC: • Seeded output (seed=42) • 200 rows to parquet • is_bad_synthetic label present
3	Synthetic bank transaction generator	data	P0	L	Implement generate_bank_transactions() — 12m daily txns: PSP settlements, supplier deb AC: • Multiple categories (psp_settlement, supplier_payment, advertising) • Running balanc
4	Synthetic PSP transaction generator	data	P0	L	Implement generate_psp_transactions() — order-level with refunds/chargebacks, settlemen AC: • Status enum (paid/refunded/chargeback) • Settlement lag T+1 to T+7 • Bad merchan
5	Abstract DataSource interface	ingestion	P0	S	Create base.py ABC with load(), validate(), load_and_validate() methods. AC: • ABC enforces load/validate • Logging on validation pass/fail
6	Bank & PSP data loaders	ingestion	P0	M	BankTransactionSource and PSPTransactionSource with schema validation. AC: • Required column checks • Null rate checks • Date range validation (>= 180 days)
7	Cross-source reconciliation	ingestion, policy	P0	M	reconcile_bank_psp() — compare bank PSP credits vs PSP net settled per merchant. AC: • Delta computed correctly • Flag RECON_FAIL if > 20% • Returns structured dict

Sprint 2: Feature Engineering & Credit Policy

Goal: 15-feature pipeline + full champion policy producing decisions with reason codes.

#	Title	Labels	Pri	Size	Description & Acceptance Criteria
8	Feature pipeline: Group A (Cashflow)	features	P0	M	Implement monthly_net_cashflow_avg_6m, revenue_volatility_90d, net_cashflow_coverag AC: • 4 features computed per merchant • Pessimistic defaults applied • Unit tests for edge
9	Feature pipeline: Group B+C (Settlement, Refunds)	features	P0	M	settlement_delay_p95, settlement_delay_median, supplier_pay_punctuality, refund_rate_lt AC: • 6 features computed • PSP data correctly filtered • Defaults for missing data
10	Feature pipeline: Group D+E (Concentration, Operational)	features	P0	M	platform_concentration (HHI), seasonality_index, ad_spend_ratio_3m, bank_psp_recon_de AC: • 5 features computed • HHI correctly calculated • Recon delta matches reconciliation r
11	Missing data handler	features	P1	S	Pessimistic defaults dict + apply_defaults() + check_data_quality() with 40% alert threshold AC: • All 15 features have defaults • NaN values filled • Portfolio-level quality report
12	Hard knock-out rules (H1–H6)	policy	P0	M	Implement _check_knockouts() — trading history, KvK, recon, balance, sanctions, GMV. AC: • 6 rules with correct thresholds from config • Returns reason codes • Unit tests per rul
13	Scored gates (S1–S8) & decision logic	policy	P0	M	Implement _score_gates() — green/amber/red scoring, inverted gates, decision thresholds AC: • 8 gates evaluated • Red >= 3 DECLINE; Red >= 2 MANUAL_REVIEW • Reason cod
14	Max amount, pricing band & full policy	policy	P0	M	credit_policy() orchestrator — knockouts → gates → amount (3-way cap) → pricing (A/B/C) AC: • Year-1 cap at EUR 25k • 0.33 debt-service ratio • Pricing bands match thresholds • F

Sprint 3: Shadow ML & Decisioning Engine

Goal: Shadow model trained + decision engine producing full envelopes logged to DuckDB.

#	Title	Labels	Pri	Size	Description & Acceptance Criteria
15	Shadow ML: training pipeline	ml	P0	L	prepare_training_data() + train_shadow_model() — GBM with 5-fold CV, AUC + Brier, featu AC: • Model pickled to models/ • CV metrics printed • Feature importances dict • Clear 'sha
16	Shadow ML: scoring & batch	ml	P0	M	score_merchant() for single merchant + batch_score() for portfolio. AC: • Returns P(default) 0–1 • Handles missing model gracefully (returns None) • Uses san
17	Decision engine: envelope assembly	decisioning	P0	M	make_decision() — combine policy result + ML score + explanations + ML narrative. AC: • All envelope fields populated • ML score flagged as informational • requires_credit_of
18	Decision audit log (DuckDB)	decisioning, monitoring	P0	M	_log_decision() — persist every decision to DuckDB with full feature vector as JSON. AC: • decision_log table auto-created • All fields stored • Queryable via SQL
19	Consent logging module	ingestion, regulatory	P1	M	log_consent_event() + get_active_consents() — DuckDB consent_log table. AC: • GRANTED/REVOKED events logged • Expiry tracked • active consents queryable
20	End-to-end pipeline orchestration	infra	P0	L	run_pipeline.py — load → features → train → decide → report summary. AC: • Runs end-to-end without errors • Prints decision summary • Example decision envelo

Sprint 4: Monitoring, Demo UI & Documentation

Goal: PSI drift monitoring, Streamlit demo, model card, fairness analysis — demo-ready MVP.

#	Title	Labels	Pri	Size	Description & Acceptance Criteria
21	PSI drift monitoring	monitoring	P1	M	compute_psi() + run_drift_report() — per-feature PSI with stable/moderate/significant statu AC: • PSI formula correct • Thresholds: <0.1 stable, 0.1–0.2 moderate, >0.2 drift • Report a
22	Portfolio metrics module	monitoring	P1	M	compute_portfolio_metrics() — approval rate, decline reasons, ML score by decision, pricin AC: • Reads from decision log • Handles empty data • Returns structured dict
23	Fairness / adverse impact analysis	monitoring, regulatory	P1	M	adverse_impact_by_sector() + adverse_impact_by_platform_count() — approval rate analy AC: • Flags sectors with approval < 80% of overall • Documents HHI bias • Returns DataFr
24	Streamlit demo application	demo	P1	L	Merchant selector → features → decision card → explanations → ML score → dashboard. AC: • 5-minute demo flow works • Traffic-light feature display • Decision card with pricing
25	Model card & documentation	ml, regulatory	P1	S	models/model_card.md — type, target, features, limitations, ethics, update policy. AC: • All sections filled • Limitations clearly stated • 'Shadow only' prominent
26	Final integration test & packaging	infra	P0	M	End-to-end test: synthetic_gen → run_pipeline → Streamlit. Package as zip with README AC: • Fresh install works (pip install + 3 commands) • All 200 merchants processed • No er